

Numerical Bilinear Interpolation for Image Rescaling

MATH 307: Numerical Analysis

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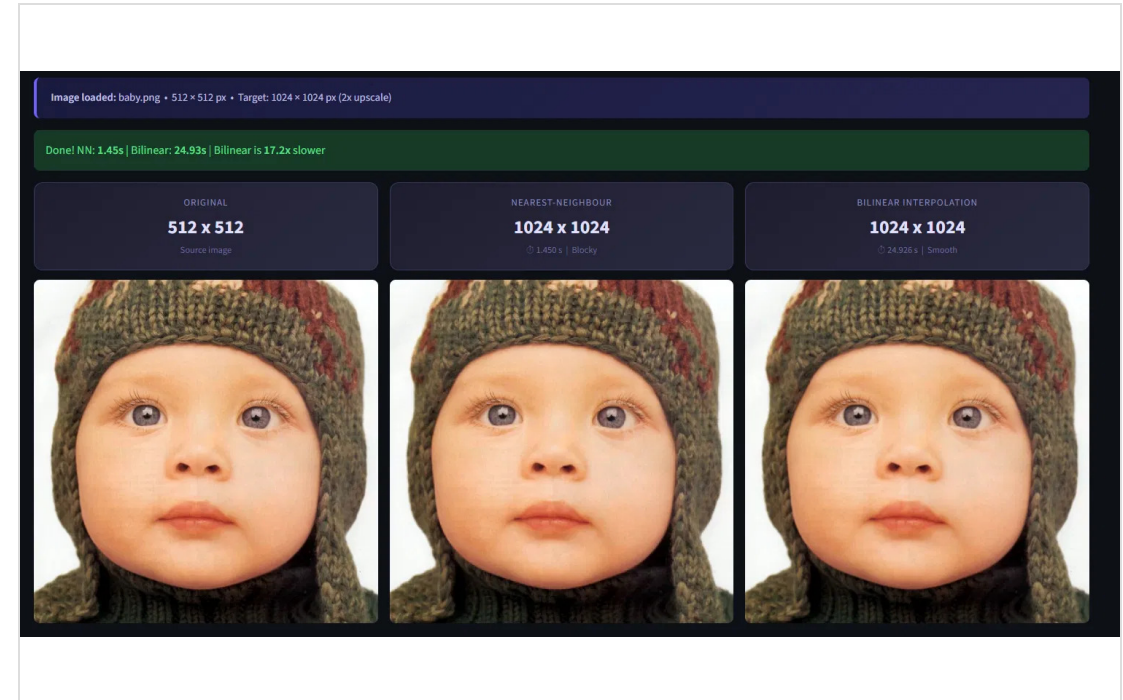
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Introduction & Goal

Digital images are inherently discrete grids. When we perform geometric operations like rescaling, we often map to non-integer coordinates.

- **The Problem:** Standard grids cannot resolve sub-pixel locations without interpolation.
- **The Goal:** Implement and compare numerical methods (Nearest-Neighbor vs. Bilinear) for smooth upscaling.
- **The Solution:** A production-grade microservices architecture featuring a FastAPI backend and Streamlit frontend.



Mathematical Formulation

Bilinear interpolation extends 1D linear splines into two dimensions. The process estimates the intensity $f(x, y)$ by weighting the four nearest integer-grid neighbors.

$$f(x, y) \approx (1-dx)(1-dy)f(x_1, y_1) + dx(1-dy)f(x_2, y_1) + (1-dx)dyf(x_1, y_2) + dx dy f(x_2, y_2)$$

- 1. Inverse Mapping:** Calculate fractional source coordinates $x = u / s_x$ and $y = v / s_y$.
- 2. Distance Weights:** Define $dx = x - \text{floor}(x)$ and $dy = y - \text{floor}(y)$ as the offsets within the pixel cell.
- 3. Sequential Interpolation:** The formula is equivalent to performing 1D Lagrange interpolation twice along the x-axis, then once along the y-axis.

Methodology & Tools

The project employs a decoupled microservices architecture to separate numerical computation from the user interface.

Technology	Role in System
FastAPI & NumPy	Backend engine for vectorized numerical processing.
Streamlit	Interactive frontend for real-time visualization.
Docker	Containerization for reproducible deployment.
Hugging Face	Cloud hosting for the public live demo.



Figure 2: The Streamlit interactive interface deployed on Hugging Face.

Nearest-Neighbor vs. Bilinear

Nearest-Neighbor (Baseline)

Assigns the value of the single closest pixel. While computationally efficient, it leads to significant **aliasing** and "blocky" artifacts, especially visible in diagonal lines and curves.

Bilinear Interpolation

Computes a weighted average of the four surrounding pixels. This creates a **continuous surface**, resulting in smooth gradients and a much higher perceived visual quality for upscaled images.

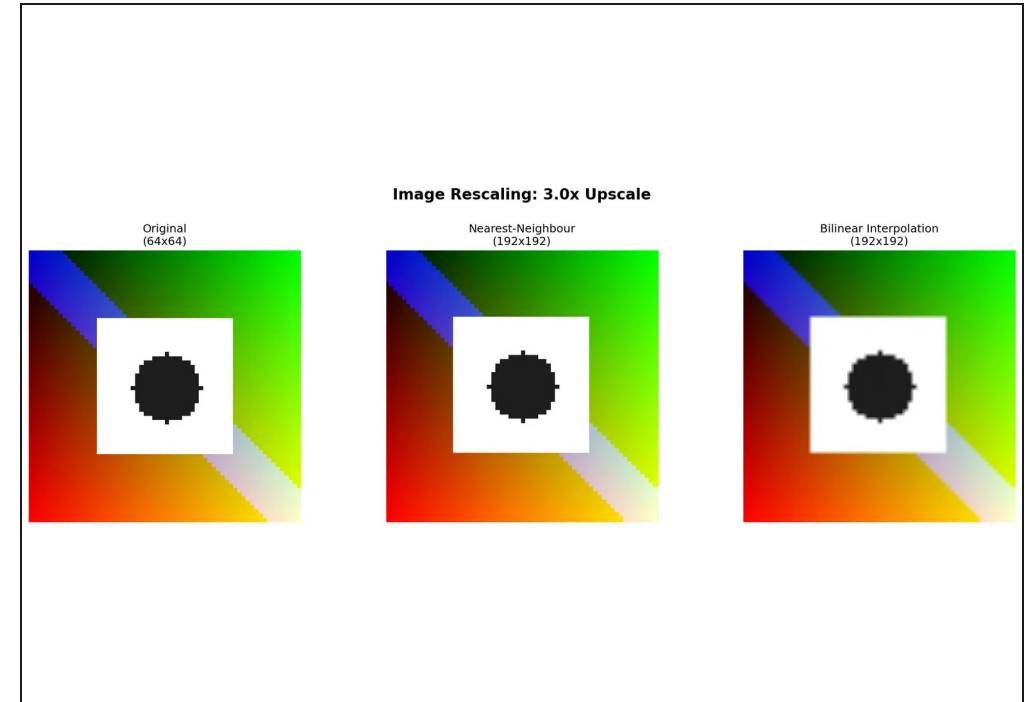


Figure 2: Visual comparison showing blocky artifacts in Nearest-Neighbor vs. smooth gradients in Bilinear Interpolation.

Experimental Results

Test Case 1: Synthetic Validation

A controlled $2 \times 2 \rightarrow 8 \times 8$ upscaling test was performed to verify mathematical correctness. The bilinear method produced perfect color blending across the grid.

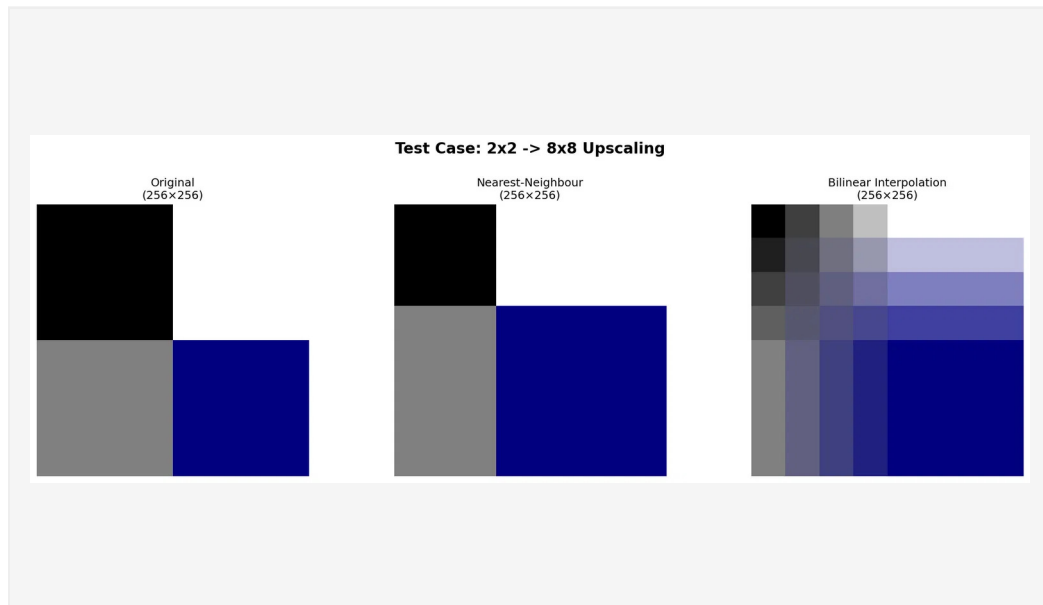


Figure 1: Mathematical validation on a synthetic color grid.

Test Case 2: Real-World Upscaling

Upscaling a 512×512 photograph to 2048×2048 . Bilinear interpolation demonstrably outperforms nearest-neighbor by eliminating blocky artifacts.

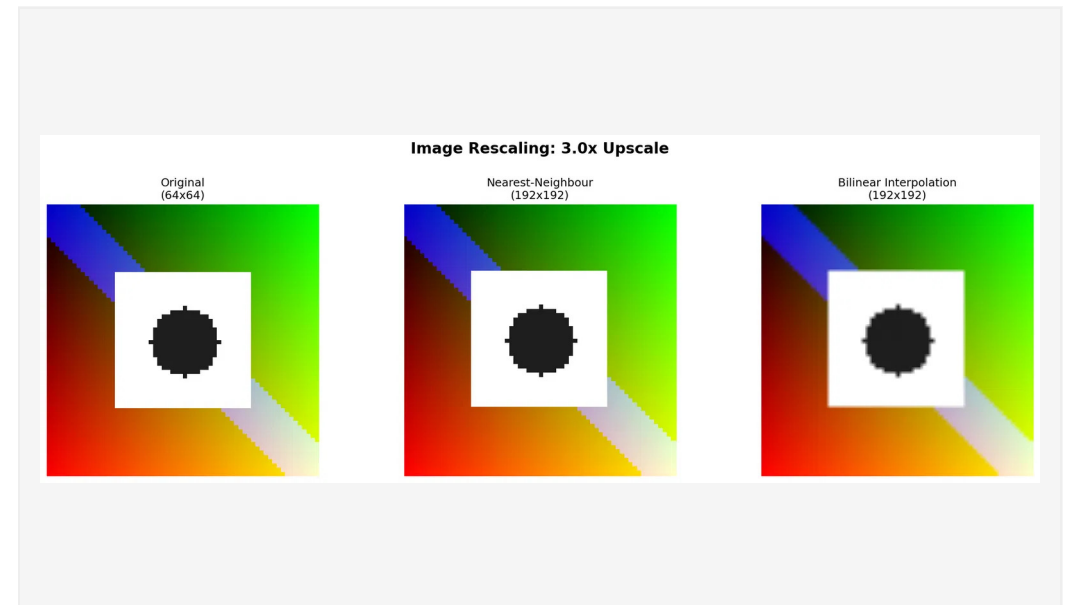


Figure 2: Side-by-side comparison of NN vs. Bilinear interpolation.

Live Demo & UI Features

The system is deployed as a fully interactive web application, allowing for real-time validation of the numerical algorithms.

Interactive Zoom Inspector

Compare raw pixels side-by-side with interpolated results using a detail-view tool.

Real-time Comparison

Toggle between Nearest-Neighbor and Bilinear methods to see immediate visual differences.

Performance Metrics

The UI reports backend processing time and numerical error metrics (MSE, PSNR).

Publicly hosted on Hugging Face Spaces via Docker containerization.



Work Division & Conclusion

Team Responsibilities

Member	Key Contributions
Mostafa Hisham	Backend Engine (FastAPI), Numerical Implementation (NumPy), Mathematical Derivation, Cloud Deployment.
Nour-Eldin Mohamed	Frontend UI (Streamlit), Dockerization, DevOps, Cloud Deployment, Numerical Implementation (NumPy), Mathematical Derivation.

Final Takeaways

- Classical numerical methods like Lagrange interpolation remain foundational to modern digital image processing.
- Bilinear interpolation provides a superior balance between computational speed and visual reconstruction quality.
- The microservices approach demonstrates how mathematical algorithms can be delivered as scalable web solutions.

Thank you for your attention.